UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
COURSE CODE	:	**CSC429**
EXAMINATION	:	**FEBRUARY 2026**
TIME	:	**3 HOURS**

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts : PART A (10 Questions)
   PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of :
   i) the Question Paper
   ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

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This examination paper consists of 6 printed pages.

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PART A

QUESTION 1

a) Name the inventor of the integrated circuit.

(1 mark)

b) In which computer generation was the integrated circuit first introduced?

(1 mark)

c) List **THREE (3)** main components of an integrated circuit.

(3 marks)

QUESTION 2

a) Discuss the computer interface that is available during the third and fourth computer generations.

(4 marks)

b) In your opinion, what is the leading cause that changed the computer interface in the third to the fourth computer generations?

(1 mark)

QUESTION 3

Question 3 is based on Figure 1. For a 16-bit computer system, assume that one character, including space, is equal to one byte and a word is equal to two bytes.

CSC429 is awesome!

Figure 1

a) What is a block?

(1 mark)

b) What is the smallest addressable unit?

(1 mark)

c) Identify the number of:
   i) byte.
   ii) bit.
   iii) word.

(3 marks)

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QUESTION 4

Solve the following equation of $-56_8 + A3_H$ using 1's and 2's complement method. Show your complete workflow.

(5 marks)

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QUESTION 5

a) Convert -27.15625_{10} to a floating-point number in IEEE single precision format in hexadecimal value form. Show your complete workflow.

(5 marks)

b) Convert $41E18000_h$ in IEEE single-precision floating-point number to decimal value form. Show your workflow.

(5 marks)

QUESTION 6

a) Discuss the whole operation of the machine cycle and its respective responsible computer components.

(8 marks)

b) Elaborate on what will happen in the machine cycle if the main memory is corrupted.

(2 marks)

QUESTION 7

Question 7 is based on Figure 2.

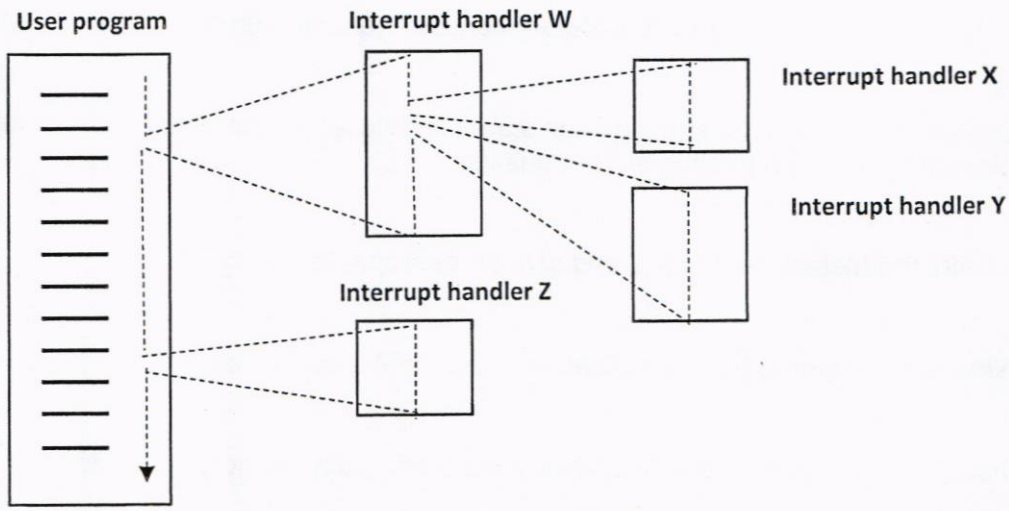


Figure 2: Nested interrupts

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a) Define interrupt.

(2 marks)

b) Identify the interrupt that has the highest priority with relevant justification

(2 marks)

c) Identify the interrupt that has the lowest priority with relevant justification

(2 marks)

d) Rank the sequence of the interrupt process completed if the computer system applies the defined priority concept to handle multiple interrupts.

(2 marks)

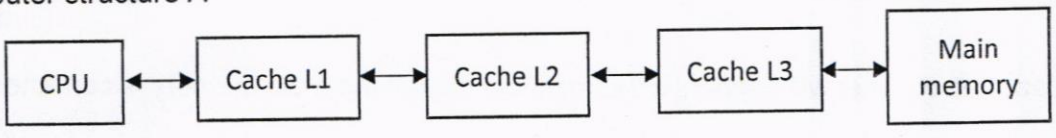
e) Rank the sequence of the interrupt process completed if the computer system applies the disable nested interrupt concept to handle multiple interrupts.

(2 marks)

QUESTION 8

Question 8 is based on Figure 3.

Computer structure A



Computer structure B

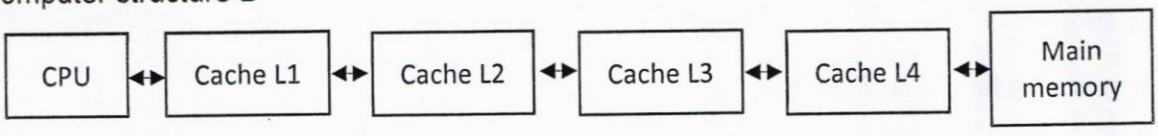


Figure 3: Different computer architectures

a) Explain which computer structure can store the data faster compared to the other. Provide relevant justification to support your answer.

(2 marks)

b) Identify the fastest cache level and give an appropriate reason.

(2 marks)

c) What is the main benefit for the cache to use SRAM technology?

(2 marks)

d) Discuss **TWO (2)** methods to update cache and main memory contents to ensure data consistency.

(4 marks)

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QUESTION 9

a) When is an I/O module installed?

(2 marks)

b) What will happen if the I/O module is not installed?

(2 marks)

c) Explain **ONE (1)** category of I/O module

(2 marks)

d) What does it mean by 'hot connection'?

(1 mark)

e) List **THREE (3)** common interfaces to external devices.

(3 marks)

QUESTION 10

a) Complete the taxonomy of parallel processor architecture illustrated in Figure 4.

(6 marks)

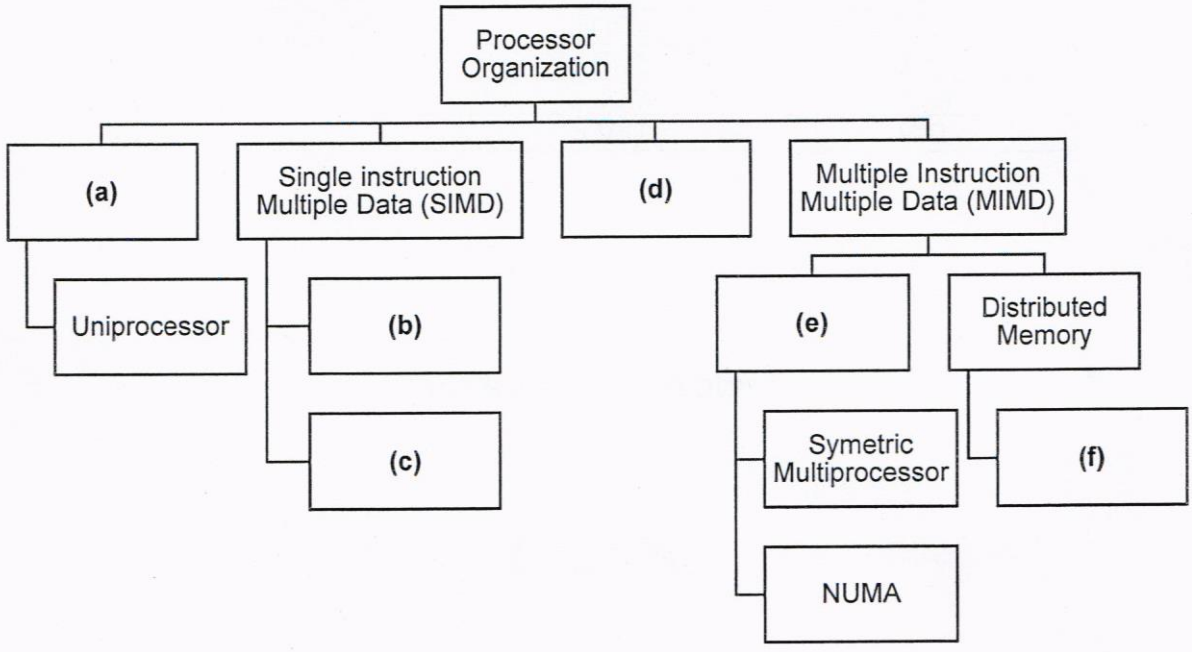


Figure 4: Taxonomy of Parallel Processor Architecture

b) Discuss **TWO (2)** disadvantages of symmetric multiprocessor (SMP).

(4 marks)

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PART B

QUESTION 1

Question 1 is based on Figure 5 and Table 1.

$$z = A + C^2 / B * D - 3A$$

Figure 5

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 5 using instructions for zero-address, one-address, and three-address. The available instructions are provided in Table 1. The registers labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register z.

(20 marks)

Table 1: Machine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	SUB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	

END OF QUESTION PAPER

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